

Chapter 2

Parametrisation of the Libor Market Model

In this chapter we study different parametrisations of the Libor market model. After some general considerations we discuss the issue of (quasi) time-shift homogeneity and introduce correlation functions and correlation structures in this context. In particular, we present a systematic approach for deriving low parametric correlation structures with nice properties, which allow for efficient and stable calibration procedures developed in Chapter 3.

2.1 General Volatility Structures

From now on we consider the Libor Market Model with respect to a tenor structure $0 < T_1 < T_2 < \dots < T_n$ in the *spot Libor measure* P^* ,

$$dL_i = \sum_{j=m(t)}^i \frac{\delta_j L_i L_j}{1 + \delta_j L_j} \gamma_i^\top \gamma_j dt + L_i \gamma_i^\top dW^*, \quad (2.1)$$

where $(W^*(t) \mid 0 \leq t \leq T_{n-1})$ is a standard d -dimensional Wiener process under the measure P^* with $d, 1 \leq d < n$, being the number of driving factors and $m(t) := \min\{m : T_m \geq t\}$ denoting the next reset date at time t . See for details Section 1.2.1

For convenience we define the *absolute or scalar volatility process* σ of the forward Libor rate L_i by

$$t \rightarrow \sigma_i(t) := |\gamma_i(t)| = \sqrt{\sum_{k=1}^d \gamma_{i,k}^2(t)}, \quad 0 \leq t \leq T_i, \quad 1 \leq i < n, \quad (2.2)$$

and *correlation structure* ρ by

$$t \rightarrow \rho_{ij}(t) := \frac{\gamma_i(t)^\top \gamma_j(t)}{|\gamma_i(t)| |\gamma_j(t)|}, \quad 0 \leq t \leq \min(T_i, T_j), \quad 1 \leq i, j < n. \quad (2.3)$$

It should be noted that the terms volatility and correlation here should be interpreted as local or instantaneous volatility and correlation. It is clear

that for any deterministic possibly time dependent d -dimensional orthogonal matrix function Q , i.e., $Q^\top Q = QQ^\top = I_d$ with I_d being the d -dimensional identity matrix, the volatility structure $\tilde{\gamma}$ defined by

$$\tilde{\gamma}_{i,k} := \sum_{l=1}^d \gamma_{i,l} Q_{lk} \quad (2.4)$$

yields the same scalar volatility and correlation structure via (2.2), (2.3). So the volatility structure γ is not completely determined by specification of the structures σ and ρ alone. However, we have the following proposition.

PROPOSITION 2.1

Consider a deterministic matrix valued map

$$Q : [0, T_{n-1}] \ni t \rightarrow Q(t) \in \mathbb{R}^{d \times d},$$

such that for each t the matrix $Q(t)$ is orthogonal, i.e., $Q^\top Q = QQ^\top = I_d$. Let us assume further the weak restriction that Q is Lebesgue measurable and bounded. Then, if the process W is a d -dimensional standard Brownian motion, $\tilde{W}$ defined by

$$\tilde{W}_t = \int_0^t Q(u) dW_u$$

is again a d -dimensional standard Brownian motion.

PROOF Since a zero-mean Gaussian process is uniquely determined by its covariance function, this follows from the fact that $\tilde{W}$ is a Gaussian process with $E\tilde{W}_t \equiv 0$, and

$$\begin{aligned} E\tilde{W}_t \tilde{W}_s^\top &= E \int_0^t Q(u) dW_u \left(\int_0^s Q(u) dW_u \right)^\top \\ &= \int_0^{\min(t,s)} I_d du = \min(t,s) I_d, \end{aligned}$$

which is the covariance function of d -dimensional standard Brownian motion. $\square$

COROLLARY 2.1

If two volatility structures γ and $\tilde{\gamma}$ are related by (2.4), their corresponding scalar volatility and correlation structures coincide and their corresponding Libor processes are identical in distribution. In this sense, γ and $\tilde{\gamma}$ may be regarded as **equivalent** volatility structures.

Economically, only the distribution of the Libor process is relevant. Therefore, by Proposition 2.1 and Corollary 2.1, the scalar volatility structure σ

and correlation structure ρ are the basic economical objects which determine the Libor market model and, loosely speaking, given these objects one may choose (almost) any convenient deterministic volatility structure γ such that (2.2) and (2.3) hold.

2.2 (Quasi) Time-Shift Homogeneous Models

2.2.1 Correlation Structures from Correlation Functions

From an economical point of view it makes sense to have some sort of time shift homogeneity in the Libor market model. Therefore it is natural to consider scalar volatility and correlation structures of the following special type,

$$\begin{aligned}\sigma_i(t) &:= c_i g(T_i - t), & 0 \leq t \leq T_i, \quad c_i > 0, \\ \rho_{ij}(t) &:= \varrho(T_i - t, T_j - t), & 0 \leq t \leq \min(T_i, T_j), \quad 1 \leq i, j < n,\end{aligned}\quad (2.5)$$

for a non-negative function $g : [0, \infty) \ni u \rightarrow \mathbb{R}_+$ and a function $\varrho : [0, \infty) \times [0, \infty) \ni (u, v) \rightarrow [-1, 1]$, which satisfies the following conditions:

- (i) $\varrho(u, v) = \varrho(v, u)$, for $u, v \geq 0$ (symmetry),
- (ii) $\varrho(u, u) = 1$, $|\varrho(u, v)| \leq 1$, for $u, v \geq 0$,
- (iii) For any number $K \in \mathbb{N}$, any set $\{u_1, \dots, u_K\} \subset \mathbb{R}_+^K$ and $\{v_1, \dots, v_K\} \subset \mathbb{R}^K$, it holds (non-negativity)

$$\sum_{k,l=1}^M \varrho(u_k, u_l) v_k v_l \geq 0.$$

A function ϱ which satisfies (i), (ii), and (iii), is called a *correlation function*.

If in (2.5) the volatility coefficients c_i are taken to be identical, i.e., $c_i = c_1$ for all i , we say that volatility structure as well as the corresponding Libor market model is *time shift homogeneous*. If $c_i \neq c_1$ we speak of a quasi time-shift homogeneous structure and model. The reason for including the coefficients c_i is that in certain applications pure time shift homogeneity of the market model may be too restrictive for calibration purposes, as we will discuss in Chapter 3.

2.2.2 Finitely Decomposable Correlation Functions

Let us consider for an integer d a unit-vector function $\eta : [0, \infty) \ni u \rightarrow \mathbb{R}^d$, hence $|\eta(\cdot)| \equiv 1$. Such a function is easily obtained by normalizing some

arbitrary $\mathbb{R}^d$ -vector function. Then, η defines a correlation function via

$$\varrho(u, v) = \eta^\top(u) \eta(v). \quad (2.6)$$

It is clear that for any sequence $0 < u_1 < \dots < u_q$, $[\rho(u_i, u_j)]_{1 \leq i, j \leq q}$ is a $q \times q$ correlation matrix with rank not larger than $\min(q, d)$. We will say that a correlation function which satisfies (2.6) for some d and η is *finitely decomposable*.

If for a finitely decomposable correlation function d and η are known, we can write down immediately a d -factor volatility structure consistent with (2.5):

$$\gamma_{i,k}(t) = c_{ig}(T_i - t)\eta(T_i - t), \quad 1 \leq i < n, \quad 1 \leq k \leq d. \quad (2.7)$$

REMARK 2.1 The class of finitely decomposable correlation functions is rather special as, for instance, it is not difficult to see that the correlation function $\varrho(u, v) := \exp(-|u - v|)$ does not belong to this class. $\square$

2.2.3 Ratio Correlation Structures and Functions

The following proposition, due to Curnow and Dunnet (1962) provides a basis for systematic generation of economically sensible correlation functions and structures for the Libor market model.

PROPOSITION 2.2

Let (b_k) , $k = 1, \dots, K$, be an arbitrary non-decreasing sequence. Then the matrix ρ defined by

$$\rho_{kl} := \frac{b_k}{b_l}, \quad \rho_{lk} := \rho_{kl}, \quad 1 \leq k \leq l \leq K, \quad (2.8)$$

is a correlation matrix. Moreover, we have a Cholesky decomposition $\rho = \eta\eta^\top$ with

$$\eta_{kl} := \frac{a_l}{b_k} 1_{l \leq k}, \quad 1 \leq k, l \leq K$$

where the sequence (a_l) is defined by $a_l := \sqrt{b_l^2 - b_{l-1}^2}$, $1 \leq l \leq K$ and $b_0 := 0$. If, in addition, the sequence (b_k) is strictly increasing, the correlation structure (2.8) has full rank.

PROOF Let Z_p , $p = 1, \dots, K$, be standard normally distributed independent real random variables and consider the random variables

$$Y_k := \sum_{p=1}^k a_p Z_p. \quad (2.9)$$

Then, for $k \leq l$ the covariance between Y_k and Y_l is given by

$$\text{Cov}(Y_k, Y_l) = \sum_{p=1}^k a_p^2 = b_k^2$$

and so the correlation is given by $\text{Cor}(Y_k, Y_l) = b_k/b_l$. Hence, (2.8) defines a correlation matrix indeed. The Cholesky decomposition can be verified straightforwardly. If (b_k) is strictly increasing we have $\eta_{kk} > 0$ for $1 \leq k \leq K$. So then the lower triangular matrix η has full rank and thus ρ also. $\square$

If we have a non-monotonic positive sequence b , we can first rearrange b to a non-decreasing sequence $\tilde{b}$ by a permutation κ , $\tilde{b}_j := b_{\kappa(j)}$, and then consider the correlation matrix $\tilde{\rho}$ obtained by $\tilde{b}$ via (2.8). Hence, for $1 \leq k, l \leq K$ we have the correlation matrix $\tilde{\rho}_{kl} = \min(\tilde{b}_k, \tilde{b}_l) / \max(\tilde{b}_k, \tilde{b}_l)$, and so $\rho_{kl} := \tilde{\rho}_{\kappa^{-1}(k), \kappa^{-1}(l)} = \min(b_k, b_l) / \max(b_k, b_l)$ is a correlation matrix as well.

COROLLARY 2.2

For any possibly non-monotonic positive sequence (b_k) we have that ρ defined by

$$\rho := \frac{\min(b_k, b_l)}{\max(b_k, b_l)}, \quad 1 \leq k, l \leq K,$$

is a correlation matrix.

COROLLARY 2.3

Consider a fixed but arbitrary positive function $\beta : [0, \infty) \rightarrow \mathbb{R}_{++}$. Then, the function

$$\varrho(u, v) = \frac{\min(\beta(u), \beta(v))}{\max(\beta(u), \beta(v))}, \quad u, v \geq 0, \quad (2.10)$$

is a correlation function.

PROOF The conditions (i) and (ii) are trivially satisfied. To prove the non-negativity condition (iii), take an arbitrary $K \in \mathbb{N}$ and a set of non-negative numbers $\{u_1, \dots, u_K\}$. Then apply Corollary 2.2 to the sequence (b_k) with $b_k := \beta(u_k)$ for $1 \leq k \leq K$. $\square$

Let us now take an increasing function $\beta : [0, \infty) \rightarrow \mathbb{R}_{++}$ and consider the correlation structure (2.5) for the correlation function (2.10). Hence,

$$\varrho_{ij}(t) := \frac{\min[\beta(T_i - t), \beta(T_j - t)]}{\max[\beta(T_i - t), \beta(T_j - t)]}, \quad t \leq \min(T_i, T_j), \quad 1 \leq i, j < n. \quad (2.11)$$

Since β is increasing, the following decomposition is clear from Proposition 2.2. For fixed t , $0 \leq t \leq T_{n-1}$, we have

$$\varrho_{ij}(t) = \sum_{k=m(t)}^{n-1} \eta_{ik}(t) \eta_{jk}(t), \quad m(t) \leq i, j < n,$$

with

$$\eta_{ik}(t) := \frac{\sqrt{\beta^2(T_k - t) - 1_{m(t) < k} \beta^2(T_{k-1} - t)}}{\beta(T_i - t)} 1_{m(t) \leq k \leq i}.$$

We thus obtain a Libor market model (2.1) consistent with (2.5) for correlation structure (2.11) by taking $d = n - 1$ and volatility structure

$$\gamma_{ik}(t) = c_i g(T_i - t) \eta_{ik}(t), \quad m(t) \leq i < n, \quad 1 \leq k < n. \quad (2.12)$$

REMARK 2.2

(i) Note that in general the correlation function (2.10) is not finitely decomposable in the sense of (2.6). Therefore, the decomposing vector function $\eta_i := (\eta_{ik})_{1 \leq k < n}^\top$ in (2.12) will in general not be time shift homogeneous, i.e., of the form $\eta(T_i - t)$ as in (2.7).

(ii) Since the function β in (2.11) is assumed to be strictly increasing, the correlation matrix $\varrho(t)$ has rank $n - 1$ for $t < T_1$. Therefore we need in (2.1) an $(n - 1)$ -dimensional Brownian motion, i.e., we need to take $d = n - 1$.

(iii) The increasingness of β has a sound economical interpretation. Namely, that the (instantaneous) correlation between forward Libors L_i and L_{i+k} decreases with k . $\square$

2.2.4 LMM with Piece-wise Constant Volatility Structure

The Hull-White Parametrisation

In Hull and White (2000) a parametrisation of the Libor market model is proposed where basically the scalar volatility and correlation functions only depend on the number of reset dates between the actual time and maturities of forward Libors.

Let $\rho^{(0)} := [\rho_{ij}^{(0)}]_{0 \leq i, j < n-1}$ be a fixed correlation matrix of rank d and $\Lambda := [\Lambda_i]_{0 \leq i < n-1}$ be a fixed vector with non-negative entries. Then the Hull-White parametrisation comes down to the following choice of scalar volatility and correlation functions,

$$\begin{aligned} \sigma_i(t) &:= \Lambda_{i-m(t)}, & 0 \leq t \leq T_i, \\ \rho_{ij}(t) &:= \rho_{i-m(t), j-m(t)}^{(0)}, & 0 \leq t \leq \min(T_i, T_j), \quad 1 \leq i, j < n. \end{aligned} \quad (2.13)$$

Let us take any set of decomposing unit vectors $\{e_i^{(0)} \in \mathbb{R}^d : 0 \leq i < n - 1\}$ with $\rho_{ij}^{(0)} = \left(e_i^{(0)}\right)^\top e_j^{(0)}$, $0 \leq i, j < n - 1$, and set $e_i(t) := e_{i-m(t)}^{(0)}$, $1 \leq i < n$.

Due to Proposition 2.1 and Corollary 2.1 the dynamics of the Hull-White model may be represented by SDE (2.1), with volatility structure γ given by

$$\gamma_i(t) := \sigma_i(t)e_i(t) = \Lambda_{i-m(t)}e_{i-m(t)}^{(0)}. \quad (2.14)$$

REMARK 2.3 For an *equidistant* tenor structure, i.e., $\delta_i = \delta_1 =: \delta$ for $1 \leq i < n$, the Hull-White volatility structure is via (2.2) and (2.3) consistent with a special version of the general (quasi) time-shift homogeneous volatility and correlation structure (2.5). Indeed, take the piecewise constant function g with $g(u) := \Lambda_{k-1}$ for k , $1 \leq k < n$, such that $(k-1)\delta \leq u < k\delta$, together with a correlation function ϱ , defined by $\varrho(u, v) := \rho_{k-1, l-1}^{(0)}$ for k, l , $1 \leq k, l < n$ such that $(k-1)\delta \leq u < k\delta$, $(l-1)\delta \leq v < l\delta$, and then take $c_i := 1$, $1 \leq i < n$. In particular, the Hull-White correlation function ϱ is decomposable since we have

$$\varrho(u, v) = \rho_{k-1, l-1}^{(0)} = \left(e_{k-1}^{(0)} \right)^\top e_{l-1}^{(0)} =: e(u)^\top e(v).$$

□

Caplet pricing

From the structure of the Hull-White model it follows that the system of Black volatilities $\{\sigma_i^B : 1 \leq i < n\}$ consistent with model prices of caplets satisfies

$$\begin{aligned} (\sigma_i^B)^2 T_i &= \int_0^{T_i} \sigma_i^2(t) dt \\ &= \sum_{k=1}^i \Lambda_{i-k}^2 \delta_{k-1} = \sum_{k=0}^{i-1} \Lambda_k^2 \delta_{i-k-1}, \quad 1 \leq i < n, \end{aligned} \quad (2.15)$$

with $\delta_0 := T_1$. In fact, the relationship (2.15) opens up the possibility of perfectly calibrating the scalar vector Λ to a system of market (for instance ATM) caplet volatilities, say $\{\hat{\sigma}_i^B : 1 \leq i < n\}$. Indeed, we can construct the vector Λ by induction: For $i = 1$ in (2.15) we get $\Lambda_0 = \hat{\sigma}_1^B$, and, if Λ_i , $0 \leq i < m$ are known, we obtain Λ_m by taking $i = m + 1$ in (2.15),

$$(\hat{\sigma}_{m+1}^B)^2 T_{m+1} - \sum_{k=0}^{m-1} \Lambda_k^2 \delta_{m-k} = \Lambda_m^2 \delta_0. \quad (2.16)$$

REMARK 2.4 It is important to note that due to (2.16) exact caplet fitting in the Hull-White model is only possible when the market caplet volatilities are such that

$$(\hat{\sigma}_{m+1}^B)^2 T_{m+1} - \sum_{k=0}^{m-1} \Lambda_k^2 \delta_{m-k} \geq 0, \quad (2.17)$$

for $1 \leq m < n - 1$. For an equidistant tenor structure this comes down to

$$(\hat{\sigma}_{m+1}^B)^2 T_{m+1} - (\hat{\sigma}_m^B)^2 T_m \geq 0, \quad 1 \leq m < n - 1. \quad (2.18)$$

So for an exact fit the caplet volatilities may not decrease too fast with maturity in fact. Unfortunately, usually (2.17) and respectively (2.18) are violated in practice; see Section 3.5. Thus, in practice perfect matching of caplets with the Hull-White model is usually not possible. $\square$

2.2.5 Modified Hull-White Volatility Structure

The Hull-White volatility structure (2.14) has the following important features: (i) The structure is pure time shift homogeneous when the tenor structure is equidistant (see Remark 2.3). (ii) For a non-equidistant but not too irregular tenor structure the volatilities can still be regarded as “nearly” time shift homogeneous. (iii) The pure or “nearly” time shift homogeneous correlation structure can be decomposed canonically into volatility factors which are also pure or “nearly” time shift homogeneous and vice versa. However, as noted in Remark 2.4 and illustrated in Section 3.5, a drawback of this model is the breakdown of recursive calibration of Λ 's to caplets. On the other hand, while exact caplet fitting of the Λ 's is usually impossible, a least squares calibration of Λ 's is likely to be unstable due to the large number of fitting parameters, e.g., $n \approx 40$. Therefore, instead of Λ 's one could take a scalar volatility structure as in (2.5), where the function g is (parsimoniously) parametrized in some suitable way. An important benefit of such scalar volatility structure is the possibility of using a smooth function g , hence smooth scalar volatility $|\gamma_i(t)|$, for instance, by choosing a nice smooth functional form of g , depending on some set of parameters. Of course, while smoothness of volatility norms can thus be retained, there is some price we have to pay. Namely, we need to include the coefficients c_i which are necessary to match the caplet vols $\hat{\sigma}_i^B$ via

$$(\hat{\sigma}_i^B)^2 T_i = \int_0^{T_i} |\gamma_i(t)|^2 dt = c_i^2 \int_0^{T_i} g^2(s) ds. \quad (2.19)$$

Note that for any particular choice of g the caplets may be matched perfectly by choosing c_i according to (2.19). However, some arbitrary choice of g would take the volatility structure far from time shift homogeneity, which might be economically less desirable. Therefore, if one would like to keep the model as close to time shift homogeneity as possible, a way to go would be, for instance, determination of a function g in some nice parametric class $\{g_\alpha : \alpha \in A\}$, such that

$$\sum_{i=1}^{n-1} \left((\hat{\sigma}_i^B)^2 T_i - c_i^2 \int_0^{T_i} g_\alpha^2(s) ds \right)^2 \rightarrow \min_{\alpha \in A, c > 0}, \quad (2.20)$$

and then determine c_i by (2.19).

The considerations above suggest to combine the Hull-White structure (2.14) with the structure (2.5) into the alternative volatility structure,

$$\begin{aligned}\gamma_i(t) &= c_i g(T_i - t) e_{i-m(t)}^{(0)}, \quad 0 \leq t \leq \min(T_i, T_j), \quad 1 \leq i, j < n, \\ \rho_{kl}^{(0)} &= \left(e_k^{(0)} \right)^\top e_l^{(0)}, \quad 0 \leq k, l < n-1.\end{aligned}\quad (2.21)$$

Thus, the structure (2.21), which can be seen as a Modified Hull-White volatility structure, contains the scalar volatility structure from (2.5) and the correlation structure from (2.14). In fact, the volatility structure (2.21) equipped with a suitably parametrised volatility function g (see (2.25) in Section 2.2.6) and a correlation matrix $\rho^{(0)}$ as developed in Section 2.3.4 will be our favorite structure for applications.

REMARK 2.5 *Non-standard forward Libors*

In practice it may happen that one needs to price a product for which the underlying forward rates do not fit with the presently standard tenor structure. For such a situation we here give a simple pragmatic way for modelling such non-standard forward Libors similar as in Brigo & Mercurio (2001a).

Note that from (1.18) we have

$$dW^{(i+1)} = dW^{(i)} + \frac{\delta_i L_i}{1 + \delta_i L_i} \gamma_i dt, \quad (2.22)$$

where $W^{(i+1)}$ and $W^{(i)}$ are standard Brownian motions under P_{i+1} and P_i , respectively. Consider the non-standard forward Libor $L_{i-1}^{\alpha, \beta}$ with respect to forward period $[T_{i-1}, T_i]$, where $T_{i-1}^\alpha := (1 - \alpha)T_{i-1} + \alpha T_i$, $T_i^\beta := (1 - \beta)T_i + \beta T_{i+1}$, and $0 \leq \alpha, \beta \leq 1$. We next introduce an intermediate standard Brownian motion $W_\beta^{(i)}$ under an intermediate measure P_i^β , connected with expiry T_i^β , by a drift interpolation,

$$dW_\beta^{(i)} := dW^{(i)} + \frac{(T_i^\beta - T_i)L_i}{1 + (T_i^\beta - T_i)L_i} \gamma_i dt. \quad (2.23)$$

Note that $\beta = 1$ yields (2.22) and $\beta = 0$ an identity. Then, for the dynamics of $L_{i-1}^{\alpha, \beta}$, where $\alpha \neq 0 \vee \beta \neq 1$, we postulate

$$\begin{aligned}dL_{i-1}^{\alpha, \beta} &= L_{i-1}^{\alpha, \beta} (\gamma_{i-1}^\alpha)^\top dW_\beta^{(i)} \\ &= \frac{(T_i^\beta - T_i)L_i L_{i-1}^{\alpha, \beta}}{1 + (T_i^\beta - T_i)L_i} (\gamma_{i-1}^\alpha)^\top \gamma_i dt + L_{i-1}^{\alpha, \beta} (\gamma_{i-1}^\alpha)^\top dW^{(i)},\end{aligned}\quad (2.24)$$

where the intermediate deterministic volatility γ_{i-1}^α is given by

$$\begin{aligned}\gamma_{i-1}^\alpha(t) &:= ((1 - \alpha)c_{i-1} + \alpha c_i) g((1 - \alpha)T_{i-1} + \alpha T_i - t) e_{i-1-m(t)}^{(0)}, \\ 0 \leq \alpha < 1, \quad 2 \leq i < n,\end{aligned}$$

and thus extends structure (2.21). In this approach, two non-standard Libors $L_{i-1}^{\alpha,\beta}$ and $L_{j-1}^{\alpha_1,\beta_1}$, with $0 \leq \alpha, \beta \leq 1$ and $0 \leq \alpha_1, \beta_1 \leq 1$, are perfectly correlated for $i = j$. Finally we note that the interpolation (2.23) differs slightly from the interpolation in Brigo & Mercurio (2001a) with respect to the denominator in the second term. In fact, by interpolation (2.23) it is guaranteed that the by (2.24) extended Libor model is automatically arbitrage-free. $\square$

2.2.6 Parametric Scalar Volatility Function

Particularly for calibration purposes, we prefer to use in (2.21) a scalar volatility function g proposed by Rebonato (1999), which is of the following parametric form,

$$g(s) = g_{a,b,g_\infty}(s) := g_\infty + (1 - g_\infty + as)e^{-bs}, \quad a, b, g_\infty > 0. \quad (2.25)$$

In fact, (2.25) is obtained as a constant plus a linear combination of the first two (orthogonal) Laguerre functions $e^{-s/2}$ and $(s-1)e^{-s/2}$ in $[0, \infty)$, properly scaled.

From a modelling point of view, the parametric function g_{a,b,g_∞} is able to reflect frequently observed “hump shape” behaviour of forward Libor volatilities, before reaching their maturity. See Figure 2.1 for a typical example.

Technically, (2.25) has the advantage that integrals of the form

$$\int_0^{T_k} g_{a,b,g_\infty}(T_i - s) g_{a,b,g_\infty}(T_j - s) ds, \quad T_k \leq \min(T_i, T_j), \quad (2.26)$$

which will play an important role in the approximation of swaptions, can be evaluated in closed form.

REMARK 2.6 The explicit expression for the integral (2.26) is horribly long and for this reason not given here. However, such an expression can be easily produced with a formula processor like Derive, Maple, or Mathematica, and then, for instance, the result can be copied into some relevant C++ implementation. $\square$

2.3 Parametrisation of Correlation Structures

2.3.1 A Disadvantage of Low Factor Models

It is known that low factor models have intrinsic problems to match (instantaneous) correlations between forward Libors realistically, see, e.g., Rebonato

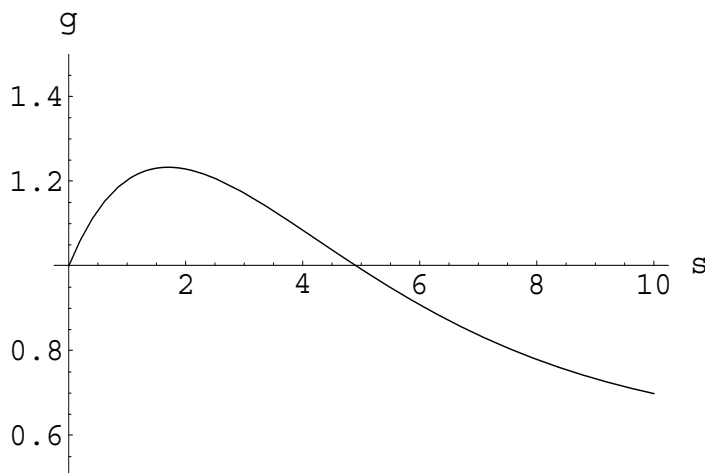


FIGURE 2.1: A typical volatility function: $s \rightarrow g_{0.5,0.4,0.6}(s)$

(1996), Schoenmakers & Coffey (1998). Let us illustrate this fact by considering a two factor Libor model with volatility structure of type (2.21), where

$$\gamma_i(t) = c_i g(T - t_i) e_{i-m(t)}^{(0)}, \quad e_k^{(0)} \in \mathbb{R}^2, \quad 1 \leq i < n, \quad (2.27)$$

and $e_k^{(0)}$, $0 \leq k < n-1$, are unit vectors which determine the initial instantaneous correlations $\rho_{kl}^{(0)} = \left(e_k^{(0)}\right)^\top e_l^{(0)}$.

Let us consider the forward Libor return correlations in the first period after $t = 0$. Since the correlation matrix $\rho^{(0)}$ has rank two, we may write

$$\frac{\gamma_i(0+)^\top \gamma_j(0+)}{|\gamma_i(0+)| |\gamma_j(0+)|} = \rho_{i-1,j-1}^{(0)} = \left(e_{i-1}^{(0)}\right)^\top e_{j-1}^{(0)} = \cos(\phi_{i-1} - \phi_{j-1}), \quad (2.28)$$

for some set of angles $\phi_0, \dots, \phi_{n-2}$ with $\phi_0 := 0$.

We now consider an artificial example. Take $n = 20$ and suppose that the market tells us the correlations $\rho_{0,j-1}^{(0)}$ behave like $\rho_{0,j-1}^{(0)} = 18/(17+j)$, thus falling down from 1 to 0.5. Then, if we calibrate this two-factor model, i.e., the ϕ_k , to these correlations it is easily seen that, as an immediate consequence, the model correlations $\rho_{j-1,18}^{(0)}$ are given by $\rho_{j-1,18}^{(0)} = \frac{9}{17+j} + \frac{\sqrt{3}}{2} \sqrt{1 - \left(\frac{17}{18} + \frac{j}{18}\right)^{-2}}$; see Figure 2.2. However, the behaviour of the correlations $\rho_{j-1,18}^{(0)}$ in figure (2.2) is clearly *not* consistent with expected real behaviour in the market which should look more or less the same as $\rho_{0,j-1}^{(0)}$, mirrored at $j = 10$. Of course the situation will be better when we increase the number of factors but a two factor example reveals this problem most clearly. In this respect we note that in Brigo and Mercurio (2001, 2001a)

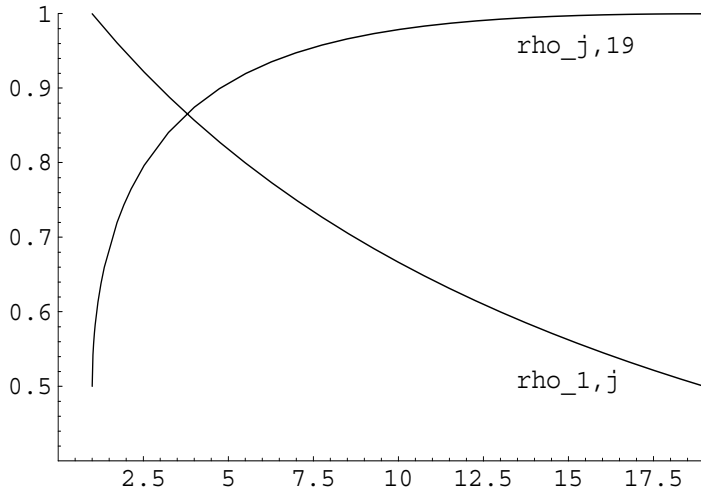


FIGURE 2.2: Example of two factor correlations

calibration of the cosine structure (2.28) is extensively tested and their conclusions concerning the behaviour of implied correlations more or less confirm the difficulties sketched above.

REMARK 2.7 It should be noted that the above considerations are not affected by the particular choice of the scalar volatility function in (2.21), since the instantaneous correlations are completely determined by the unit vectors $e_k^{(0)}$ alone. $\square$

As a solution for the intrinsic low factor problem we consider in the next section multi-factor models based on a natural framework of ratio correlation structures (Section 2.2.3), earlier proposed in Schoenmakers & Coffey (1998) and further developed in Schoenmakers & Coffey (2000,2003), in which correlation behaviour observed in practice is nicely reflected. Correlation structures in this framework involve basically the same number of parameters as the general rank two correlation structure (2.28).

2.3.2 Semi-parametric Framework for Libor Correlations

We consider a multi-factor Libor market model (2.1) consistent with volatility structure (2.21) with $d = n - 1$, and where $e_i^{(0)}$ are unit vectors which determine a particular initial correlation structure as introduced below.

A Semi-parametric Libor Correlation Structure is by definition a ma-

trix $\rho^{(0)}$ such that

$$\rho_{kl}^{(0)} = \left(e_k^{(0)} \right)^\top e_l^{(0)} = \frac{\min(b_k, b_l)}{\max(b_k, b_l)}, \quad 0 \leq k, l < n-1, \quad (2.29)$$

for a strictly increasing sequence $b = (b_0, \dots, b_{n-2})$ which is such that in addition

$$k \longrightarrow \frac{b_k}{b_{k+1}} \text{ is strictly increasing.} \quad (2.30)$$

Since a correlation structure (2.29) essentially involves $\mathcal{O}(n)$ parameters, whereas a general or *non-parametric* correlation matrix would require $\mathcal{O}(n^2)$ entries, we may call (2.29) a *semi-parametric* correlation structure.

The increasingness of the sequence b implies that the instantaneous model correlations $Cor(\Delta L_i, \Delta L_{i+k})$ decrease with k for fixed i , and the additional assumption (2.30) implies that $Cor(\Delta L_i, \Delta L_{i+k})$ increases with i for fixed k . Both implications reflect natural behaviour of forward rates. For instance, on the one hand, the correlation between a two and a ten year forward is expected to be lower than the correlation between a two and five year forward whereas, on the other hand, the correlation between a six and ten year forward is expected to be higher than the correlation between a two and six year forward.

Because of Corollary 2.1 we may assume without restriction that $e^{(0)}$ is lower triangular. In particular, due to the imposed correlation structure (2.29) and Proposition (2.2), we can take $e_{k,l}^{(0)} = 1_{0 \leq l \leq k} \frac{1}{b_k} \sqrt{b_l^2 - b_{l-1}^2}$ with $b_{-1} := 0$. Hence, with $g_i(t) := c_i g(T_i - t)$ we may write

$$\begin{aligned} \Delta L_i &= \dots \Delta t + L_i g_i \sum_{l=0}^{i-m(t)} e_{i-m(t),l}^{(0)} \Delta W_l^* \\ &= \dots \Delta t + L_i g_i \sum_{l=0}^{i-m(t)} \frac{1}{b_{i-m(t)}} \sqrt{b_l^2 - b_{l-1}^2} \Delta W_l^* \\ &= \dots \Delta t + L_i g_i \sqrt{1 - \frac{b_{i-m(t)-1}^2}{b_{i-m(t)}^2}} \Delta W_i^* + \frac{L_i g_i}{L_{i-1} g_{i-1}} \frac{b_{i-m(t)-1}}{b_{i-m(t)}} \Delta L_{i-1}. \end{aligned} \quad (2.31)$$

The interpretation of (2.31) is clear: The risky part of a forward Libor increment ΔL_i at time t , $t < T_{i-1}$, is a linear combination of the forward increment ΔL_{i-1} and an independent random shock ΔW_i^* , with coefficients determined by $L_{i-1}(t)$, $L_i(t)$, $g_{i-1}(t)$, $g_i(t)$, $b_{i-m(t)-1}$, and $b_{i-m(t)}$, respectively. We emphasize that decomposition (2.31) is due to the special structure of the correlation matrix (2.29) and does not hold for a general (non-parametric) correlation structure.

2.3.3 Representation Theorems

The next theorem provides a representation for the sequence b in the semi-parametric structure (2.29), where the non-linear constraints on b are transformed into simple constraints in $\mathbb{R}_+^{n-2}$.

THEOREM 2.1

For every non-decreasing sequence $b := (b_1, \dots, b_m)$ with $b_1 := 1$ and b such that $i \rightarrow b_i/b_{i+1}$ non-decreasing, there exists a sequence of non-negative numbers $\Delta_i, \Delta_i \geq 0, i = 2, \dots, m$, such that

$$b_i = \exp \left[\sum_{l=2}^m \min(l-1, i-1) \Delta_l \right], \quad 1 \leq i \leq m. \quad (2.32)$$

Conversely, for any sequence Δ_l with $\Delta_l \geq 0, l = 2, \dots, m$, (2.32) is non-decreasing, b_i/b_{i+1} is non-decreasing, $b_1 = 1$. Moreover, if b is given by (2.32) then

$$\begin{aligned} \Delta_i &= 2 \ln b_i - \ln b_{i-1} - \ln b_{i+1}, \quad 2 \leq i < m, \\ \Delta_m &= \ln b_m - \ln b_{m-1}; \end{aligned}$$

hence the representation (2.32) is unique.

If in addition $\Delta_m > 0$, the sequence b is strictly increasing, and, if $\Delta_i > 0, i = 2, \dots, m-1$, the sequence b_i/b_{i+1} is strictly increasing.

PROOF To exclude trivialities we may assume $m \geq 3$. Suppose we are given a non-decreasing sequence b with b_i/b_{i+1} non-decreasing. Let us define, $\xi_i := \ln b_i, 1 \leq i \leq m$. Then, $\xi_1 = 0$ since $b_1 = 1$ and ξ_i satisfies the following constraints,

$$\begin{aligned} \xi_i &\leq \xi_{i+1}, \quad 1 \leq i \leq m-1 \\ \xi_{i-1} + \xi_{i+1} &\leq 2\xi_i, \quad 2 \leq i \leq m-1. \end{aligned} \quad (2.33)$$

We now introduce the new variables,

$$\begin{aligned} \Delta_i &:= 2\xi_i - \xi_{i-1} - \xi_{i+1} = \xi_i - \xi_{i-1} - (\xi_{i+1} - \xi_i) \geq 0, \quad 2 \leq i \leq m-1, \\ \Delta_m &:= \xi_2 - \sum_{i=2}^{m-1} \Delta_i = \xi_m - \xi_{m-1} \geq 0, \end{aligned} \quad (2.34)$$

and thus it follows straightforwardly that for $2 \leq i \leq m$,

$$\xi_2 - \sum_{l=2}^{i-1} \Delta_l = \xi_i - \xi_{i-1} \geq 0, \quad (2.35)$$

with an empty sum defined to be zero. We so obtain by (2.35) for $2 \leq i \leq m$,

$$\begin{aligned}\xi_i &= \xi_i - \xi_1 = \sum_{k=2}^i \xi_k - \xi_{k-1} = \sum_{k=2}^i \left\{ \xi_2 - \sum_{l=2}^{k-1} \Delta_l \right\} \\ &= (i-1)\xi_2 - \sum_{k=3}^i \sum_{l=2}^{k-1} \Delta_l = (i-1)\xi_2 - \sum_{l=2}^{i-1} \sum_{k=l+1}^i \Delta_l \\ &= (i-1)\xi_2 - \sum_{l=2}^{i-1} (i-l)\Delta_l.\end{aligned}\tag{2.36}$$

Then via (2.34) and (2.36) we may express ξ_i , and so b_i , in the new parameters Δ_i by

$$\begin{aligned}\xi_i &= (i-1) \sum_{l=2}^m \Delta_l - \sum_{l=2}^{i-1} (i-l)\Delta_l = (i-1) \sum_{l=i}^m \Delta_l + \sum_{l=2}^{i-1} (l-1)\Delta_l \\ &= \sum_{l=2}^m \min(l-1, i-1)\Delta_l, \\ b_i &= \exp(\xi_i).\end{aligned}\tag{2.37}$$

The converse and uniqueness follow by straightforward derivation of (2.34) for $\xi_i := \ln b_i$ with a sequence (b_i) given by (2.32) for a non-negative sequence $\Delta_2, \dots, \Delta_m$.

Combining (2.34) and (2.35) yields

$$\xi_i - \xi_{i-1} = \sum_{l=i}^m \Delta_l, \quad 2 \leq i \leq m.$$

Hence, if b is given by (2.32) with $\Delta_i \geq 0$, the sequence b is over all *strictly* increasing if and only if $\Delta_m = \xi_m - \xi_{m-1} > 0$. Finally, by (2.34) the sequence b_i/b_{i+1} is strictly increasing if and only if $\Delta_i > 0$ for $i = 2, \dots, m-1$. $\square$

COROLLARY 2.4

A correlation structure of type (2.29), generated by a sequence b as in Theorem 2.1, can be represented by

$$\rho_{ij} = \exp \left[- \sum_{l=i+1}^m \min(l-i, |j-i|) \Delta_l \right], \quad 1 \leq i, j \leq m, \tag{2.38}$$

with $\Delta_i > 0$, $2 \leq i \leq m$.

REMARK 2.8 For consistency with the notation in (2.29) in connection with a Libor market model, one can take in (2.38), $m = n-1$ and set $\rho_{kl}^{(0)} := \rho_{k+1, l+1}$, $0 \leq k, l < n-1$. $\square$

As an alternative, one can also derive semi-parametric Libor correlation structures (2.29) from Corollary 2.3 by taking a function β which is, in addition to Corollary 2.3, strictly increasing and such that for each $v \geq 0$ the function $s \rightarrow \beta(s)/\beta(s+v)$ is strictly increasing as well. Indeed, then for any increasing sequence u_k , $0 \leq k < n-1$, in $\mathbb{R}_+$, the sequence $b_k := \beta(u_k)$, $0 \leq k < n-1$, defines a Libor correlation structure (2.29). In this context we have the following representation theorem for the correlation generating function β .

THEOREM 2.2

Let $\beta : [0, \infty) \rightarrow \mathbb{R}_+$ be two times continuously differentiable in $(0, \infty)$ with (i): $\beta(0) = \beta(0+) = 1$, (ii): β is non-decreasing, and (iii): for each $v > 0$ the function $s \rightarrow \beta(s)/\beta(s+v)$ is non-decreasing.

Then, there exists a non-negative continuous function η in $(0, \infty)$, with (iv): $\int_1^\infty \eta(z)dz < \infty$, (v): $\int_{0+}^1 dy \int_y^\infty \eta(z)dz < \infty$, and a constant $C \geq 0$, such that

$$\beta(s) = \exp\left[\int_0^s dy \int_y^\infty \eta(z)dz + Cs\right]. \quad (2.39)$$

Conversely, let $C \geq 0$ and η in $(0, \infty)$ be non-negative, continuous, and such that it holds (iv), (v). Then, (2.39) defines a two times differentiable function in $(0, \infty)$ which satisfies (i), (ii), (iii), and $\beta(0+) = 1$.

If, in addition, η is strictly positive, then both β and $s \rightarrow \beta(s)/\beta(s+v)$ are strictly increasing for each $v \geq 0$.

PROOF Let us begin with the converse statement. If C and η are as stated, β is well defined by (2.39). Moreover, by direct verification it follows that (i), (ii), and (iii) hold. Indeed, we have

$$\beta'(s) = \beta(s)\left(\int_s^\infty \eta(z)dz + C\right) \geq 0, \quad \text{and} \quad (2.40)$$

$$\begin{aligned} \frac{d}{du} \frac{\beta(u)}{\beta(u+s)} &= \frac{d}{du} \exp\left[\int_0^u dy \int_y^\infty \eta(z)dz + Cu\right. \\ &\quad \left.- \int_0^{u+s} dy \int_y^\infty \eta(z)dz - C(u+s)\right] \\ &= \frac{\beta(u)}{\beta(u+s)} \left(\int_u^\infty \eta(z)dz - \int_{u+s}^\infty \eta(z)dz\right) \\ &= \frac{\beta(u)}{\beta(u+s)} \int_u^{u+s} \eta(z)dz \geq 0. \end{aligned} \quad (2.41)$$

Now suppose that β is two times continuously differentiable in $(0, \infty)$ and such that (i), (ii), and (iii) hold. Then by differentiability of β , we have for

any $s, u > 0$,

$$\beta(u+s)\beta'(u) - \beta(u)\beta'(u+s) \geq 0,$$

due to (iii). Hence, for any $s, u > 0$,

$$\frac{\beta'(u)}{\beta(u)} - \frac{\beta'(u+s)}{\beta(u+s)} \geq 0,$$

so β'/β is both non-negative and non-increasing. Hence, we may define $C := \lim_{u \rightarrow \infty} \beta'(u)/\beta(u)$ with $C \geq 0$. Since β is two times differentiable in $(0, \infty)$, we can define a function η on $(0, \infty)$ by

$$\eta(u) := -\frac{d}{du} \frac{\beta'(u)}{\beta(u)} = -\frac{d^2}{du^2} \ln \beta(u),$$

which is thus non-negative. It then follows that for $u > 0$,

$$\frac{\beta'(u)}{\beta(u)} = \int_u^\infty \eta(z) dz + C,$$

where in particular the r.h.s. integral exists, hence (iv). Next, by integration and taking into account $\beta(0+) = 1$, we obtain

$$\ln \beta(s) = \int_{0+}^s du \int_u^\infty \eta(z) dz + Cs, \quad (2.42)$$

and, in particular, existence of the r.h.s. integral, hence (v), and we have (2.39).

Finally, if in addition η is strictly positive, the left-hand sides of (2.40) and (2.41) are strictly positive.

□

COROLLARY 2.5

A function β which satisfies the requirements of Theorem 2.2 induces a correlation function (2.10) of the form,

$$\begin{aligned} \rho(s, u) &= \frac{\min[\beta(s), \beta(u)]}{\max[\beta(s), \beta(u)]} \\ &= \exp \left[-C|u-s| - \left| \int_s^u dy \int_y^\infty \eta(z) dz \right| \right]. \end{aligned}$$

Example 2.1

Let us take $C := 0$ and, for some constants γ, ρ with $0 < \gamma < 1$, $\rho > 0$, the function $\eta(z) := \rho\gamma(1-\gamma)z^{\gamma-2}$ in $(0, \infty)$. Then, clearly, (iv) and (v)

of Theorem 2.2 hold and we thus obtain via (2.39) a correlation generating function,

$$\beta(s) := \exp(\rho s^\alpha).$$

So, for instance, the sequence b with $b_i := \beta(i) = \exp(\rho i^\alpha)$, $0 \leq i < n-1$, defines a correlation structure (2.29). $\square$

In fact, Theorem 2.2 and Corollary 2.5 are presented because of their nice analogy with Theorem 2.1 and Corollary 2.4. Our further study of correlation structures, however, will be exclusively based on Theorem 2.1 and Corollary 2.4, for the following reason. Theorem 2.1 establishes a one to one correspondence between any correlation generating sequence b in (2.29) and a non-negative sequence Δ , whereas in general there are infinitely many functions β in Theorem 2.2, which may generate the same sequence b .

2.3.4 Generation of Low Parametric Structures

From Corollary 2.4 we can derive conveniently various low parametric structures which all fit into the framework of Libor market models (2.21), equipped with correlation structures of the form (2.29). We give some examples.

Example 2.2

Let us take $\Delta_2 = \dots = \Delta_{m-1} =: \alpha \geq 0$ and $\Delta_m =: \beta \geq 0$. Then, (2.38) yields the correlation structure

$$\rho_{ij} = \exp \left[-|i-j| \left(\beta + \alpha \left(m - \frac{i+j+1}{2} \right) \right) \right], \quad 1 \leq i, j \leq m. \quad (2.43)$$

Note that for $\alpha = 0$ we get $\rho_{ij} = e^{-\beta|i-j|}$, a simple correlation structure frequently used in practice in spite of the, in fact, unrealistic consequence that $i \rightarrow \text{Cor}(\Delta L_i, \Delta L_{i+p})$ is *constant* rather than increasing for fixed p . Let us introduce new parameters $\rho_\infty := \rho_{1m}$ and $\eta := \alpha(m-1)(m-2)/2$, hence

$$\beta = -\frac{\alpha}{2}(m-2) - \frac{\ln \rho_\infty}{m-1} = \frac{-\eta - \ln \rho_\infty}{m-1}.$$

Then (2.43) becomes

$$\rho_{ij} = \exp \left[-\frac{|i-j|}{m-1} \left(-\ln \rho_\infty + \eta \frac{m-i-j+1}{m-2} \right) \right], \quad 0 \leq \eta \leq -\ln \rho_\infty, \quad 1 \leq i, j \leq m. \quad (2.44)$$

While the structures (2.43) and (2.44) are essentially the same, by the re-parametrization of (2.43) into (2.44) the parameter stability is improved: Relatively small movements in the b -sequence connected with (2.44), thus the (market) correlations, cause relatively small movements in the parameters ρ_∞

and η . In fact, this can be seen also by analytical comparison of the parameter sensitivities (derivatives) in (2.43) and (2.44). $\square$

The following three parametric structure is a refinement of (2.43):

Example 2.3

Suppose $m > 2$ and let Δ_i be a linear function of i for $2 \leq i \leq m-1$, with $\Delta_2 =: \alpha_1 \geq 0$, $\Delta_{m-1} =: \alpha_2 \geq 0$. Hence, for $i = 2, \dots, m-1$, we have

$$\Delta_i = \alpha_1 \frac{m-i-1}{m-3} + \alpha_2 \frac{i-2}{m-3}$$

and we further define $\Delta_m =: \beta \geq 0$. Then, from (2.38) we get by rather tedious but elementary algebra the correlation structure

$$\begin{aligned} \rho_{ij} = \exp \left[-|j-i| \left(\beta - \frac{\alpha_2}{6m-18} (i^2 + j^2 + ij - 6i - 6j - 3m^2 + 15m - 7) \right. \right. \\ \left. \left. + \frac{\alpha_1}{6m-18} (i^2 + j^2 + ij - 3mi - 3mj + 3i + 3j + 3m^2 - 6m + 2) \right) \right]. \end{aligned} \quad (2.45)$$

Note that (2.45) collapses to (2.43) for $\alpha_1 = \alpha_2 = \alpha$. As in (2.43) we now re-parameterize (2.45) by $\rho_\infty = \rho_{1m}$ which yields

$$\beta = \frac{-\ln \rho_\infty}{m-1} - \frac{\alpha_1}{6} (m-2) - \frac{\alpha_2}{3} (m-2).$$

In order to gain parameter stability as in (2.44), we set

$$\alpha_1 = \frac{6\eta_1 - 2\eta_2}{(m-1)(m-2)}, \quad \alpha_2 = \frac{4\eta_2}{(m-1)(m-2)}$$

and then (2.45) becomes

$$\begin{aligned} \rho_{ij} = \exp \left[-\frac{|j-i|}{m-1} \left(-\ln \rho_\infty \right. \right. \\ \left. \left. + \eta_1 \frac{i^2 + j^2 + ij - 3mi - 3mj + 3i + 3j + 2m^2 - m - 4}{(m-2)(m-3)} \right. \right. \\ \left. \left. - \eta_2 \frac{i^2 + j^2 + ij - mi - mj - 3i - 3j + 3m + 2}{(m-2)(m-3)} \right) \right], \end{aligned} \quad (2.46)$$

$$1 \leq i, j \leq m, \quad 3\eta_1 \geq \eta_2 \geq 0, \quad 0 \leq \eta_1 + \eta_2 \leq -\ln \rho_\infty.$$

Obviously, for $\eta_1 = \eta_2 = \eta/2$, (2.46) yields (2.44) again. Correlation structure (2.46) is used for implied calibration procedures in Chapter 3. $\square$

Example 2.4

A realistic two-parametric correlation structure. Since calibrating a three parametric structure takes longer than a two-parametric one of course, we

next present an also realistic two parameter structure obtained from (2.46) by taking $\eta_2 = 0$ (equivalently, $\alpha_2 = 0$ in (2.45)),

$$\begin{aligned} \rho_{ij} = \exp \left[-\frac{|j-i|}{m-1} \left(-\ln \rho_\infty \right. \right. \\ \left. \left. + \eta \frac{i^2 + j^2 + ij - 3mi - 3mj + 3i + 3j + 2m^2 - m - 4}{(m-2)(m-3)} \right) \right], \\ 1 \leq i, j \leq m, \quad \eta > 0, \quad 0 < \eta < -\ln \rho_\infty. \end{aligned} \quad (2.47)$$

For this correlation structure the magnitude of concavity of the sequence $\ln b_i$ in (2.29) is decreasing to zero rather than being constant like in (2.43). $\square$

Example 2.5

It is easily checked that the sequence b defined by

$$b_i = e^{\beta(i-1)^\alpha}, \quad 1 \leq i \leq m, \quad \beta > 0, \quad 0 < \alpha < 1, \quad (2.48)$$

satisfies the requirements of Theorem 2.1 and thus yields a correlation structure of the form (2.29), which was proposed in Schoenmakers & Coffey (1998) and used in Kurbanmuradov, Sabelfeld & Schoenmakers (2002) for testing Libor approximations (Chapter 6). By Theorem 2.1, the structure (2.48) has a representation (2.32) with

$$\begin{aligned} \Delta_i &= 2\beta(i-1)^\alpha - \beta(i-2)^\alpha - \beta i^\alpha, \quad 2 \leq i \leq m-1, \\ \Delta_m &= \beta(m-1)^\alpha - \beta(m-2)^\alpha, \end{aligned} \quad (2.49)$$

where $\Delta_i > 0$ for $1 \leq i \leq m$, and Δ_i is strictly decreasing for $2 \leq i \leq m-1$. By introducing $\rho_\infty := 1/b_m$, we get from (2.48)

$$\rho_{ij} = e^{\ln \rho_\infty \left| \left(\frac{i-1}{m-1} \right)^\alpha - \left(\frac{j-1}{m-1} \right)^\alpha \right|}, \quad 1 \leq i, j \leq m, \quad 0 < \alpha < 1, \quad \beta > 0. \quad (2.50)$$

In principal, (2.50) has similar properties as the structure (2.47). See further Example 2.1, where basically the same sequence (2.48) is derived from a correlation generating function. $\square$

REMARK 2.9 Let us now consider the parametrization of market correlations used by Rebonato (1999a), which has for an equidistant tenor structure the following form

$$\rho_{ij} = \rho_\infty + (1 - \rho_\infty) \exp[-|j-i|(\beta - \alpha \max(i, j))]. \quad (2.51)$$

The structure (2.51) can be seen as a perturbation of the correlation structure

$$\hat{\rho}_{ij} = \rho_\infty + (1 - \rho_\infty) \exp[-|j-i|\beta] \quad (2.52)$$

and has the desirable property that

$$i \rightarrow \rho_{i,i+p} = \rho_\infty + (1 - \rho_\infty) \exp[-p(\beta - \alpha(i + p))]$$

is *increasing* for $\alpha > 0$ and thus may produce for a given tenor structure realistic market correlations for properly chosen ρ_∞ , $\beta > 0$ and (small) $\alpha > 0$; see Rebonato (1999a). However, while (2.51) may produce realistic correlations, it should be noted that its $(\alpha, \beta, \rho_\infty)$ domain of positivity is not explicitly specified. Hence, for a particular choice of parameters it is not directly guaranteed that (2.51) defines a correlation structure indeed. In particular, it can be verified easily that (2.51) does not fit in the framework of (2.29). It is clear that all the correlation structures consistent with (2.29), in particular (2.43), (2.45), (2.48), and their equivalent representations, do not suffer for this problem as, from the way they are constructed, they are endogenously positive definite and incorporate the economically realistic properties $j \rightarrow \rho_{ij}$ is *decreasing* for $j > i$ and $i \rightarrow \rho_{i,i+p}$ is *increasing* for fixed p , for any choice of parameters *in their well specified domain*. This property makes the correlation structures consistent with (2.29) presented in this section particularly suitable for calibration purposes as search routines are thus prevented for searching in parameter regions where the matrix ρ_{ij} fails to be a correlation structure in fact. $\square$

2.3.5 Parametric Low Rank Structures

For different reasons, for example computational efficiency, it can be desirable to work with a d -factor Libor model where d is much smaller than n , the number of Libors. For example one wants to use a two or three factor model (d is two or three, respectively). As we are generally interested in parsimonious volatility structures we thus need to generate parsimonious but meaningful parametrisations of low rank correlation structures. As an example, suppose we want to construct a two factor Libor model with volatility structure of type (2.21). Since $d = 2$ we need in (2.21) a $(n - 1) \times (n - 1)$ correlation matrix $\rho^{(0)}$ of rank two. Now observe that such a correlation matrix can always be written as

$$\begin{aligned} \rho_{kl}^{(0)} &= \left(e_k^{(0)} \right)^\top e_l^{(0)} = \cos(\phi_k - \phi_l), \quad 0 \leq k, l < n - 1, \\ e_{k1}^{(0)} &:= \cos(\phi_k) \quad e_{k2}^{(0)} := \sin(\phi_k), \end{aligned} \quad (2.53)$$

for some set of angles $\phi_0, \dots, \phi_{n-2}$ with $\phi_0 := 0$. In fact, the structure (2.53) is a special case ($d = 2$) of the general angle representation for a correlation matrix of rank d . See Rousseeuw and Molenberghs (1993), Rebonato (1999a), Rapisarda, Brigo and Mercurio (2002). The representation (2.53) has still $n - 1$ degrees of freedom and a sensible parametrisation of it comes down to an economically sensible parsimonious parametrisation of the angles ϕ_k . It is clear that such a parametrisation, for instance, in the spirit of (2.29) is not

obvious. In Rebonato (1999a) a Monte Carlo based least squares calibration of a certain rank- d angle structure (e.g., (2.53) for $d = 2$) to some fixed target (“market”) correlation matrix is proposed. Such a search procedure, however, requires in general relatively many parameters to calibrate, $O(nd)$ in fact. As explained in Chapter 3, for calibration purposes we prefer parsimonious structures and therefore we seek for a suitable way to parametrise a low rank correlation matrix with only a few number of parameters. A natural tool for this is Principal Component Analysis (PCA).

Principal Component Analysis

Let us consider a full rank $(n-1) \times (n-1)$ correlation matrix ρ . Then, for any choice of the factor dimension d , we can map ρ into a rank d correlation matrix by the method of Principal Component Analysis: We take the first d principal components of ρ , i.e., corresponding to the eigenvalues $\lambda_1 > \dots > \lambda_d \geq 0$ (we assume different eigenvalues), we take unit eigenvectors $u_1, \dots, u_d$. Then we define $e_i \in R^d$, $i = 1, \dots, n-1$, by $e_i(p) = u_p(i)\sqrt{\lambda_p}$, $p = 1, \dots, d$. Next, in order to get a rank- d correlation matrix $\rho^{(d)}$ we re-normalize these vectors and so define $\tilde{e}_i \in R^d$ by $\tilde{e}_i = e_i/|e_i|$, to obtain

$$PCA(\rho; d) := \tilde{\rho}^{(d)} := (\tilde{e}_i^\top \tilde{e}_j), \quad 1 \leq i, j < n.$$

Let us start with a full rank parametric correlation structure, for example we take the structure $\rho_{ij}^{\eta_1, \eta_2, \rho_\infty}$ in (2.46). Then, by PCA one can produce for any choice of d the correlation structure

$$\tilde{\rho}_{ij}^{(\eta_1, \eta_2, \rho_\infty; d)} := PCA(\rho^{\eta_1, \eta_2, \rho_\infty}, d), \quad (2.54)$$

being a rank- d correlation structure parametrized by three parameters $\eta_1, \eta_2, \rho_\infty$.

REMARK 2.10 It should be noted that the rank- d correlation structure, resulting from the operation

$$\rho^{\eta_1, \eta_2, \rho_\infty} \rightarrow PCA(\rho^{\eta_1, \eta_2, \rho_\infty}, d),$$

usually cannot be given by some relatively simple analytic expression such as (2.46) which we have for $\rho^{\eta_1, \eta_2, \rho_\infty}$. However, within a certain calibration procedure involving the parameters $\eta_1, \eta_2, \rho_\infty$ for instance, the PCA operation can be carried out effectively by standard numerical procedures; see for example, *Numerical Recipes*, Press et al. (1992). $\square$

2.4 Some Possible Applications of Parametric Structures

2.4.1 Smoothing Historical Libor Correlations (sketch)

We assume a fixed time tenor structure $T_0 < T_1 < T_2 < \dots < T_n$ with equi-distant tenors, i.e., $\delta = T_{i+1} - T_i$ for all i , and a system of risk-free zero-coupon bonds B_i also called “present values” which mature at T_i , hence $B_i(T_i) = 1$. We further suppose that up to the present calendar date T_0 time series of historical data of the present values $B_i(t)$, $t \leq T_0$, $i = 1, \dots, n$ are available, for instance on a daily basis. These values can be obtained from the present value curve $T \rightarrow B(t, T)$, $t \leq T_0$, $T \geq T_0$, which in turn may be constructed by interpolating the available present values at each historical date $t \leq T_0$. For more details on yield curve smoothing methods, see, e.g., Fisher et al. (1995), Linton et al. (1998). So, in principle, time series of forward Libor rates are available via

$$L_i(t) = \left(\frac{B_i(t)}{B_{i+1}(t)} - 1 \right) \delta^{-1}, \quad t \leq T_0, \quad i = 1, \dots, n-1.$$

As a rough approximation of the Libor market model, by neglecting the drifts in fact, we assume that returns of the forward Libor rates may be modelled by a multi-dimensional Gaussian distribution. We thus assume

$$\frac{\Delta L_i(t)}{L_i(t)} \approx \gamma_i^\top \Delta Z \sqrt{\Delta t},$$

where $\gamma_i \in \mathbb{R}^{n-1}$ is assumed to be constant over the time interval where the time series are considered, and ΔZ is an $(n-1)$ -dimensional standard Gaussian vector, i.e., $\mathbb{E} \Delta Z_i \Delta Z_j = \delta_{ij}$. We so have

$$\mathbb{E} \frac{\Delta L_i}{L_i} \frac{\Delta L_j}{L_j} = \gamma_i^\top \gamma_j \Delta t = |\gamma_i| |\gamma_j| \rho_{ij} \Delta t. \quad (2.55)$$

In (2.55) the $|\gamma_i|$ can be identified by historical estimates of the left-hand side for $i = j$, since $\rho_{ii} = 1$. Then, in (2.55) we plug in the estimations for $|\gamma_i|$ and some parametric correlation structure, for example one of the full rank structures (2.46), (2.47), or some parametric low rank structure (2.54) obtained via Principal Component Analysis. Finally, we calibrate via least squares the parameters of this correlation structure to the left-hand side estimations of (2.55) for $i \neq j$. Resuming we propose the following program for identification of a smooth Libor return correlation structure:

- I *Construct daily time series for the different forward Libors*
- II *Estimate historically the left-hand side of (2.55) in the usual way and thus obtain for $i = j$ in particular historical estimations for the $|\gamma_i|$.*

- III *Fit the (for instance two or three) parameters of a parametric correlation structure ρ via (2.55) by standard least squares to the historical estimations obtained for $i \neq j$.*

2.4.2 Calibration to Caps and Swaptions by Using Historical Correlations (sketch)

One can take in Section 2.4.1 the historically identified instantaneous correlation structure as fixed $\rho^{(0)}$ in the volatility structure (2.21) of a Libor market model and then calibrate some parametric scalar volatility function g to the prices of swaptions via a least squares objective function proposed in Section 3.2.1. For instance, one could take the parametric form (2.25). Finally, after identifying g the coefficients c_i can be obtained by fitting the caplets via (2.19).

We note, however, that this calibration method involves historical estimation of Libor correlations which, on the one hand, may be a rather elaborate job and, on the other hand, is not really desirable in the view of implied calibration enthusiasts who follow the idea that the Libor model should be identified by exclusively using price information relevant at the calibration date (T_0). Full implied calibration methods are described next in Chapter 3.